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Irrigation and tillage management effects on solute movement

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Abstract

In developing management practices to reduce chemical leaching below the root zone, tillage and irrigation management are important considerations. Two studies were performed to evaluate the movement of bromide in tilled and non-tilled soils under sprinkler versus flood irrigation. In each study, bromide was applied either with an irrigation or presprayed to the soil surface followed by periodic soil sampling to monitor the bromide movement. Tillage was observed to reduce the mean depth of chemical penetration under both irrigation treatments and reduce the spatial variation of bromide concentration under flood irrigation. For example, after 30 days of periodic flood irrigation, 25% of the applied bromide remained in the upper 0.2 m of a tilled soil while in the companion non-tilled soil virtually no bromide remained above this depth. The most rapid bromide movement was observed in non-tilled, flood irrigated soil, particularly when the solute was added with the irrigation. We speculate that the tillage effect of reduced leaching results from the alteration of pore continuity and creation of diffusional sinks and not increased evaporative water loss in the tilled soil. The Root Zone Water Quality Model was calibrated using site-specific hydraulic property measurements and used to predict the solute movement. The model predictions were fairly accurate for the sprinkler irrigated soil but less satisfactory for the flood irrigation studies. In comparing the effect on chemical leaching of the treatments imposed, we found that tillage and the timing of the chemical application had greater impact on reducing leaching than did the method of irrigation. © 1998 Elsevier Science B.V. All rights reserved.

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1. Introduction

Irrigation and tillage management are important interacting factors in the leaching of agricultural chemicals. Tillage alters soil structural features such as pore size distribution, total macroporosity, and

macropore continuity (Benjamin, 1993), which are key elements determining the hydraulic conductivity of the soil. The irrigation technique influences water flow patterns in the soil as a consequence of the uniformity of the application and the application flux or potential head. Rapid movements of water and solutes through continuous macropores has attracted a good deal of interest in the literature (Thomas and Phillips, 1979; Beven and Germann, 1982; White,

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1985; Andreini and Steenhuis, 1990; Edwards et al., 1993; Flury et al., 1994). A common finding of this research is that regions of exceptionally high hydraulic conductivity have the potential to conduct water and solutes at a much greater rate than would occur in a homogeneous material of lower conductivity, and, in so doing, cause a bypass of much of the soil matrix. While, strictly speaking, water will always flow preferentially through paths of least resistance, the term *preferential flow* has joined the lexicon to identify flow which exceeds rates expected by classic Darcian concepts using macroscopically averaged hydraulic conductivity and fluid pressure (Beven, 1991). In addition to the difficulty of accurately characterizing the prevalence of macropores in the field, predicting macropore flow requires consideration of a combination of factors. The existence of macropores in itself does not guarantee enhanced solute movement. In terms of management factors, the literature suggests it is the interplay of tillage (influencing macropore prevalence and continuity), irrigation type and intensity (controlling soil water potential at the soil surface), and chemical placement and timing relative to the irrigation event which dictate the potential or probability of rapid solute movement short-circuiting the soil matrix (Granovsky et al., 1993; Quin et al., 1994). For example, Kanwar et al., 1985, using a high intensity rainfall simulation, reported a perhaps unexpected finding of greater leaching losses of nitrate (spray applied 18 h prior to irrigation) in a tilled soil than in non-tilled plots. Apparently the greater macropore water flow in the non-tilled soil bypassed the resident nitrate within the micropores of the soil aggregates with the result of reduced nitrate leaching. The role of soil aggregates and microporosity in reducing the movement of solutes in the macropore flow stream was demonstrated by Ahuja et al. (1995) and Heathman et al. (1995). Similarly, Jaynes et al. (1991) observed a three-fold increase in deep leaching when a tracer solute was chemigated as opposed to spray-applied prior to irrigation. In as much as solute diffusion into the liquid filled microporosity is not instantaneous, a short lag time between application and an intense irrigation event favors solute movement with the macropore flow (Insensee et al., 1990; Edwards et al., 1992).

The objective of this work is to evaluate how two methods of irrigation (sprinkler and flood) affect the

movement of a non-adsorbing solute in tilled and non-tilled soil. As an added treatment factor, we will compare the effect of chemigation versus pre-application on the solute leaching. We do not evaluate all possible treatment combinations of tillage, irrigation, and solute application timing. Our approach will be to report and integrate the results of two studies which serve to illustrate these treatment effects on solute movement. We are particularly interested in determining whether the tracer movement in non-tilled soil is retarded relative to the movement in the tilled soil, suggestive of water flow by-pass resident solute. Conversely, it could be hypothesized that the homogenizing effects of tillage reduce pore continuity and consequently reduce leaching. The importance of these mechanisms in causing greater or lesser chemical leaching should depend on the type of irrigation, hence the emphasis on the effects of the irrigation method. We conclude by determining whether the observed chemical movement could have been predicted with a recently introduced solute transport/management model, the Root Zone Water Quality Model (Ahuja et al., 1993).

2. Materials and methods

2.1. Study I: Chemigated solute movement under flood irrigation in tilled and non-tilled soil

The first experiment was conducted near Fort Collins, Colorado, on an Ulm clay loam (fine, montmorillonitic, mesic Ustollic Haplargid). The soil displays strong prismatic structure from the surface to a depth of 1.2 m with frequent large cracks and root/worm holes. Below 1.2 m, the soil profile grades into structureless form.

Two field plots of 5 m by 5 m, one tilled and the other non-tilled, were established for flood irrigation. The tilled plot was plowed to a depth of about 0.2 m and finely tilled with a two wheel rotary tiller to a depth of about 0.1 m. The non-tilled plot (non-tilled for seven consecutive years) was prepared by spraying a solution of glyphosate (*N*-phosphomethyl-glycine) to kill the vegetation. About 10 days after the herbicide application, the plot was mowed and raked carefully to remove dead plant parts. The outer perimeter of the plots as well as an inner 3 m by 3 m sampling area was

surrounded by dikes to prevent surface runoff during irrigations. A solution of analytic grade KBr was sprayed on each plot at the rate of 59.5 kg ha^{-1} (50 mmol m^{-2}). At the time of application, the plots were ponded to about 5 cm with water to simulate a chemigation input of solute.

Following solute application, the fallow plots were flood irrigated (in an identical fashion) every 7 to 10 days. Soil cores were collected 8, 30 and 76 days following the bromide application. For each sampling event, three closely spaced 2.5 cm diameter soil cores were obtained from 16 “points” in the plot. The depth of coring depended on the sampling event but was always at least 0.96 m. Following retrieval of the cores, the resultant holes were back-filled. The cores were sectioned into 0.05 or 0.10 m increments (smaller increments near the surface) and returned to the laboratory. The soil samples were split for gravimetric determination of water content and saturation paste extraction (Rhoades, 1986) for bromide content. The bromide concentration in the extracts was determined using a Lachet ion-chromatography autoanalyzer. The bromide concentration in the soil solution was then calculated from the extract concentration taking into account the mass of soil extracted, the gravimetric water content of the sample, and the volume of water used for the extraction. As a quality assurance test of the extraction and analytical methods employed, bromide free soil from the field site was treated with a known bromide mass and extracted and analyzed. Using 15 samples from three depths (0.2, 0.5 and 1.5 m) a mean recovery of $107 \pm 12\%$ was determined (not a statistically significant deviation from 100% recovery at the 0.95 confidence level).

2.2. Study II: Non-chemigated solute movement in non-tilled soil under flood versus sprinkler irrigation

The second experiment, also near Fort Collins, Colorado, was conducted on a sandy clay loam (fine loamy mixed mesic Ustollic Haplargid) with a moderate sub-angular blocky structure in the upper 2 m. Below 2 m the soil is massive. Earthworms (*Lumbricus* spp.) with resultant channels (approximately 2 mm diameter) were fairly common (estimated at 160 pores m^{-2}) in this soil above 0.3 m soil depth. No

visible cracks or biopores were observed below about 1 m. The site had not been disturbed for at least five years prior to this study.

Two 4 m by 4 m plots were prepared; each plot contained a 2.5 m by 2.5 m inner area for soil core sampling. Both plots were mowed and gently raked of fresh clippings, but neither plot was tilled. One of the plots was set up for sprinkler irrigation, while the other was diked for flood irrigation. A KBr solution was sprayed at a rate of 148.8 kg ha^{-1} to each plot 18 h prior to the first irrigation. The sprinkler irrigated plot was irrigated with a system containing eight sprinkler heads producing a discharge rate of 3 cm/h (but operated intermittently to prevent ponding) at a coefficient of uniformity $\geq 90\%$. In the flood irrigated plot, the flooding was accomplished with four 2.54 cm diameter hoses with a total discharge rate of 1.61 l s^{-1} . To prevent erosion, the discharge end of each hose was inserted into an irrigation sock. To improve uniformity of application, the hoses were shifted within the plot a few times during each irrigation. The plots were sampled four times following the solute application (5, 19, 34, and 69 days). For each sampling, 25 mm diameter cores were retrieved from eight locations in the plot. Soil cores from like depths were composited and returned to the laboratory for extraction and analysis as indicated for Study I.

To aid in determination of soil water losses by evaporation, hanging water table, draining microlysimeters (0.2 m diameter $\times$ 0.2 m length cylinders) were monitored for each study site. The lysimeters were prepared by enclosing a carefully excavated monolith of tilled or non-tilled soil in a PVC cylinder and capping the lower boundary with a porous membrane. Fine air-dried soil was used to smooth irregularities in the lower boundary of the monolith and ensure good contact with the membrane. To simulate drainage conditions of a non-confined soil, a hanging water table was attached to the outflow ports of the lysimeter. The lysimeter was installed in a 0.21 m diameter access hole with a smaller diameter bore hole extending about 2 m below the lysimeter to allow for the hanging water table vessel. The access hole for the lysimeter was lined with welded wire to support the walls and allow for easy and repeated removal of the lysimeter. The lysimeters were weighed and the drainage volumes recorded on a daily basis.

3. Results and discussion

3.1. Study I

The objective of the first study was to evaluate the effect of tillage on the movement of solute applied and followed by flood irrigation. The spatial variability of the recovered bromide from the first sampling (Fig. 1) suggests highly variable water movement in both the tilled and non-tilled soil. The bromide mass per unit area at each location was calculated by integrating the products of the bromide concentration (as mmol/m^3 of soil solution) and the volumetric water content over depth (m). Although not visually obvious, the mean values in Fig. 1(a) and (b) are nearly identical and represent 90 and 95% recovery of the applied mass/area in the tilled and the non-tilled plots, respectively. The coefficient of variation (standard deviation/mean)

of the recovered bromide mass, however, is 0.66 in the tilled soil and nearly double that (1.17) in the non-tilled soil. The large lateral variability in the bromide concentration, particularly in the non-tilled soil, suggests water flow in preferred pathways funneled the nearly uniformly applied solute into localized regions creating area of high and low concentration. The solute mass distribution in the later samplings (not shown) continue to display the localization effect, but, interestingly, there is little vertical correlation in the high concentration regions with time. This may arise from the multidimensional water flow preventing extremes in solute concentration from persisting vertically at a given coordinate in the plot. It is also possibly a sampling artifact due to the necessity of spatial separation (0.2 m) of sequential soil cores from the same “point”. If the mass localization resulted from preferential funnelling of the application pool by macropore/crack features, then solute concentration would be expected to have very short lateral correlation distances and even small spatial separation in sequential sampling would miss vertical correlation.

As a result of the mollifying effects of tillage on the variability of water and solute movement, the area averaged solute concentration profiles in the tilled and non-tilled soil are distinctly different (Fig. 2). At $T=30$ days (Fig. 2(a)), virtually no bromide is detected in the upper 0.2 m of the non-tilled soil, whereas this same region in the tilled soil contains the highest bromide concentrations. This tillage effect on the solute concentration is again observed at $T=76$ days (Fig. 2(b)) where the recovered bromide in the non-tilled soil is below 0.4 m, while in the tilled soil, the bromide appears to loiter in the tillage zone. [For clarity, confidence limits (CL) were omitted in Fig. 2. The 95% CL are wide, typically $\pm 60\%$ of the mean values shown. Despite the large concentration variability, the difference between the mean concentrations in the tilled and non-tilled soil in the upper 0.2 m at $T=30$ days and the upper 0.4 m at $T=76$ days is statistically significant (at 99% confidence level using 2-tailed t -test)].

A second major difference apparent in the bromide distributions in the tilled and non-tilled soil is the lower mass recovery in the non-tilled soil. As suggested by the relative areas under the concentration curves (Fig. 2), the mass recovery in the non-tilled soil deteriorates with time. In fact, by day 76, 58% of the

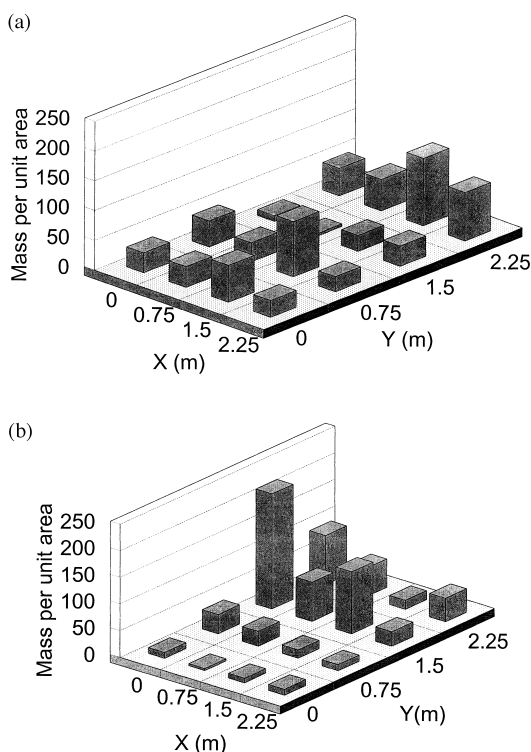


Fig. 1. Spatial distribution of bromide mass per unit area (mmol m^{-2}) recovered eight days after application in (a) tilled and (b) non-tilled soil under periodic flood irrigation (Study I).

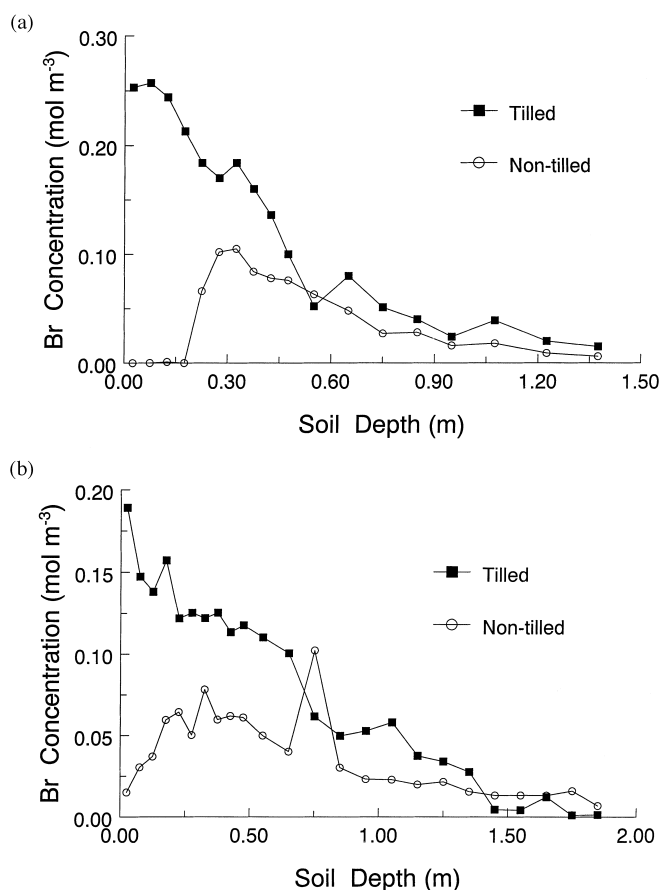


Fig. 2. Bromide concentration profiles in tilled (■) and non-tilled (○) soil after (a) 30 days and (b) 76 days under periodic flood irrigation (Study I).

applied bromide mass is unaccounted for in the non-tilled soil, compared to 25% missing in the tilled soil.

Taken together, these results suggest that tillage reduced the leaching of a chemigated, non-adsorbing solute. Tillage breaks the continuity of structural features, which, under ponding, allow rapid water movement and hence, transport of a large portion of the applied solute mass. Irrigation subsequent to the chemigation in the non-tilled soil does not appear to have bypassed the resident bromide (as might be expected if the solute had diffused into immobile water zones) but instead continued still deeper leaching. The missing bromide in the non-tilled soil is probably below the greatest sampling depth and/or resides as high concentration pockets missed in the sampling. In the tilled soil, we speculate that the dead-

end porosity created by the tillage provides a diffusion “sink” for the tracer, and hence, its concentration remains high in the tillage zone for the duration of the experiment. It could be argued that the larger near surface bromide concentration in the tilled soil arises from an increased evaporative flux due to tillage effects on hydraulic properties. This argument is not supported by the lysimeter data, however. A comparison of cumulative water loss by evaporation over a 150 day period finds less than 4% difference between the tilled and low residue non-tilled soil.

The application of the bromide tracer with the ponding infiltration provided optimum conditions for rapid transport of the solute in the non-tilled soil. In a similar experiment at the same field site, Fleming and Butters (1995) introduced a bromide tracer with

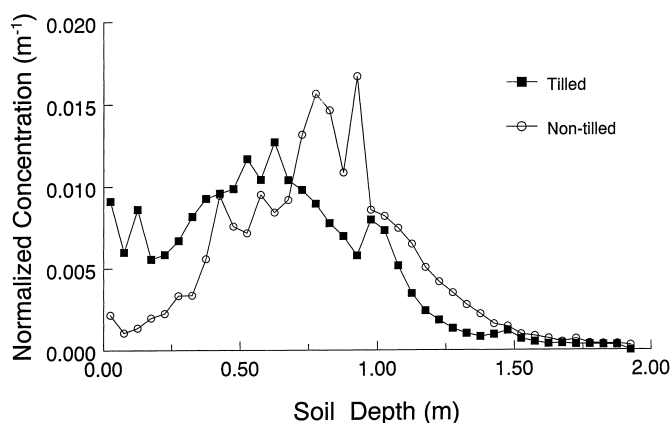


Fig. 3. Bromide concentration profiles observed by Fleming and Butters (1995) in tilled (■) and non-tilled (○) soil under periodic sprinkler irrigation (same site as Study I).

non-ponding, sprinkler irrigation to tilled and non-tilled soil. Their results (Fig. 3) show the same, though somewhat less distinct, tillage zone solute retention effect as observed in this study. Fleming and Butters monitored the progress of the bromide plumes for over a year. Even after the bromide had leached below the tillage zone, its long residence time in the tillage zone doomed it lag behind the zealously convective plume in the non-tilled soil. As was the case here, the results of Fleming and Butters did not suggest that soil water in the non-tilled soil “by-passed” resident solute and hence reduced leaching relative to tilled soil.

3.2. Study II

In the second study, the primary focus was to examine the effect of the irrigation method (sprinkler vs. flood) on the leaching of a preapplied tracer. As discussed previously, the bromide tracer was sprayed on two non-tilled plots 18 h prior to the first irrigation. As illustrated in Fig. 4, the center of mass of the bromide distributions 34 and 69 days later (Fig. 4) is nearly identical in the two plots. Greater solute dispersion in the flood irrigated plot was expected and was detected, especially in the final sampling. Under ponding irrigation, all pore sizes are potential participants in water and chemical flow, whereas the largest pores are expected to be excluded under the non-saturating sprinkler irrigation. As discussed above, water flow in the largest pores may cause rapid and far reaching convection of a portion of the applied

solute while simultaneously bypassing other regions of the soil containing solute. The result is larger variability in the solute velocity and hence greater vertical spreading of the solute plume under flood irrigation than with sprinkle irrigation. We expect that the somewhat subtle irrigation treatment effect on solute dispersion observed here with pre-applied solute would be more pronounced for the case of a chemigated solute.

During the execution of Study II, an additional experiment was underway in an adjacent plot at the same field site which produced results which can be compared with Study II. Part of this other (yet to be published) study involved evaluation of bromide movement in non-tilled soil under sprinkler irrigation but with chemigated bromide instead of a pre-application as performed here. A comparison of the bromide concentration profiles from the two experiments sampled after similar quantities of net irrigation is made in Fig. 5. The profiles are very similar, with the exception of the upper 0.15 m. It appears that pre-application of the bromide provides greater opportunity for access to diffusional sinks near the soil surface, which somewhat reduces the leaching.

4. Model prediction

As a final point, it is useful to investigate whether the bromide travel depth profiles observed in the two main studies discussed above have been predicted. To

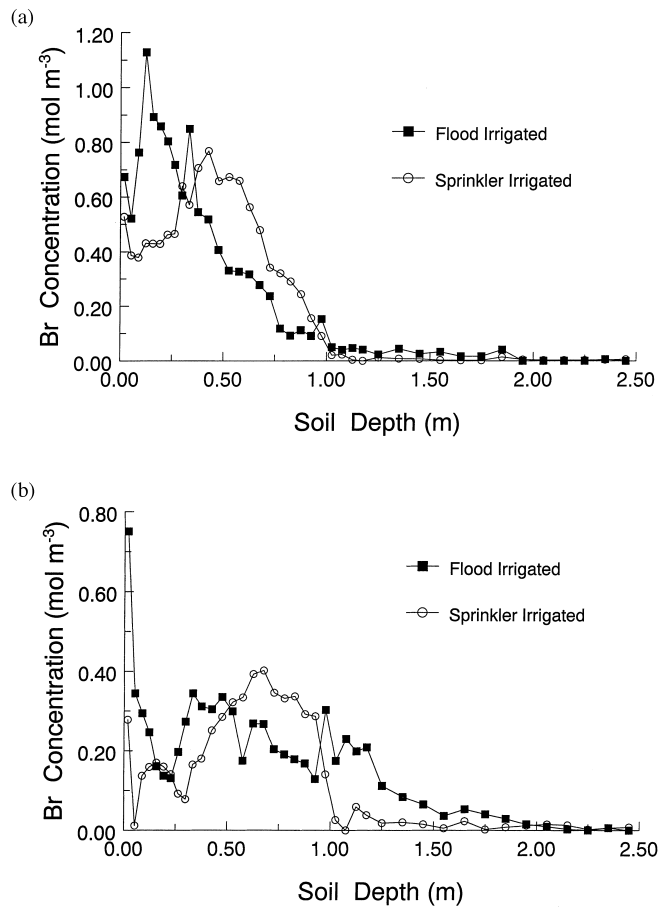


Fig. 4. Bromide concentration profiles in non-tilled soil under periodic flood (■) and sprinkler (○) irrigation after (a) 34 days and (b) 69 days (Study II). The mass recovery in these sequential samplings is 95 and 65%, respectively, in the flood irrigation plot and 97 and 53%, respectively, in the sprinkler irrigated plot.

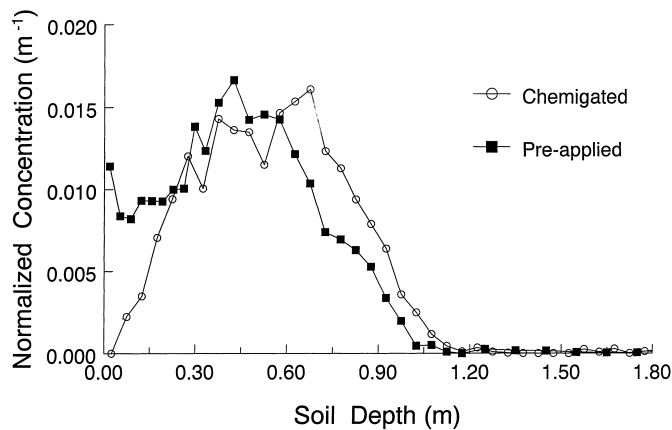


Fig. 5. Solute concentration profiles in sprinkler irrigated, non-tilled soil for pre-applied (■) versus chemigated (○) bromide after similar quantities of net applied water.

this end, we have employed the Root Zone Water Quality Model (for details of the RZWQM, see Ahuja et al., 1993), which is specifically designed to simulate water and chemical movement under varying management practices. In the RZWQM, the soil is modeled as two interacting domains: the matrix, where traditional concepts of flow and transport prevail, and the macropores, where a flow capacity is estimated using Poiseuille's law. To execute the model, we supplied the soil physical and hydraulic properties measured on site. Given the conceptual rigor of the RZWQM, the parameter estimation tasks can be formidable. The key soil hydraulic properties [$K(h)$ and $\theta(h)$] for each soil horizon to a depth of 1.8 m were determined using the instantaneous profile method as described by Green et al. (1986). Macropore numbers were estimated using tension infiltration (Watson and Luxmoore, 1986) for each horizon of the soil profile. Other important inputs included the management factors (tillage type and timing; irrigation type, timing and quantity) and meteorological data for estimation of evaporative water losses.

The RZWQM predictions of the bromide concentration profiles observed in the two studies presented here are shown in Figs. 6 and 7. The results are shown for a single sampling time but are typical of the model's predictive accuracy over the duration of the experiment. In Study I involving flood irrigation in tilled and non-tilled soil (Fig. 6), RZWQM clearly underpredicts the observed bromide movement. The

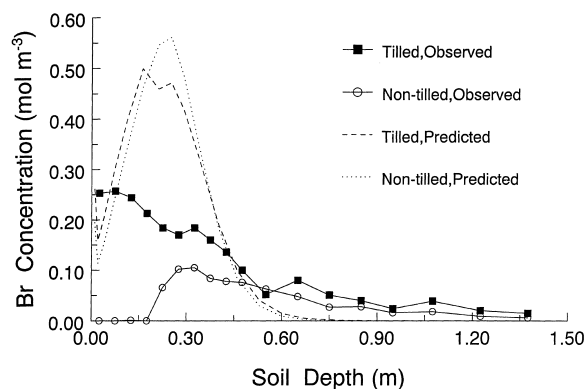


Fig. 6. RZWQM predictions of the plot average bromide concentration profiles from Study I at $t=30$ days for flood irrigated tilled soil [data (■), model (...)] and non-tilled soil [data (○), model (...)].

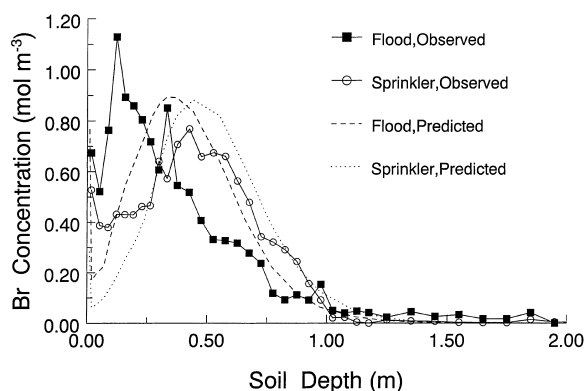


Fig. 7. RZWQM predictions of the bromide concentration profiles from Study II at $t=34$ days for sprinkler irrigated tilled soil [data (■), model (...)] and sprinkler irrigated [data (○), model (...)], non-tilled soil.

model predicts nearly identical results in the tilled and non-tilled soil. It appears as if the model is shuttling too much solute-free water in the macropore flow domain, which bypasses bromide near the soil surface and reduces the water available to leach bromide in the soil matrix. In the comparison experiment (Study II) involving flood versus sprinkler irrigation in non-tilled soil (Fig. 7), the RZWQM predictions are more accurate but appear to underestimate dispersion, particularly in the flood irrigated soil. The discrepancies in the RZWQM predictions may be an inherent failing of model description of the system and/or reflective of errors in the experimental techniques used in calibrating the model. It is clear that evolution of ever more sophisticated computer models with difficult-to-measure parameters will continue the challenge to experimentalists and probably outpace the field testing of such models.

5. Summary and conclusions

In a comparison of bromide transport in tilled and non-tilled soil under flood or sprinkler irrigation, bromide leaching was most rapid in flood irrigated, non-tilled soil with chemigation. Bromide leaching losses were reduced by sprinkler irrigation, tillage, and pre-application of the solute. The tillage zone physical retention of bromide relative to non-tilled soil is probably a consequence of reduced pore continuity

and bromide diffusion into low mobility pore water. Similarly, diffusion into immobile pore water explains the reduced leaching of pre-applied solute relative to chemigated solute. The mean displacement of bromide was similar under flood and sprinkler irrigation, but, as expected, there was a noticeable difference in dispersion. Surface ponding provides for participation of all pore sizes, which allows both rapid solute movement and by pass of resident solute. Consequently the solute profiles under flood irrigation show greater bromide concentration both ahead and behind the profile under sprinkler irrigation. A water flow/solute transport management model (RZWQM) was fairly accurate in its prediction of solute concentration profiles under sprinkler irrigation but faltered for flood irrigation cases in both tilled and non-tilled soil.

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